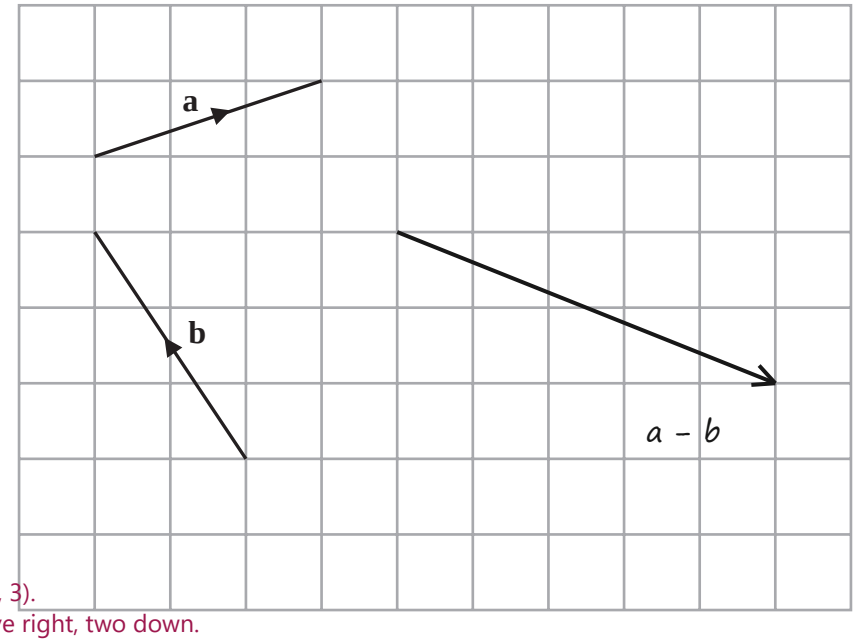
10 The vector **a** and the vector **b** are shown on the grid.



1 mark
a = (3, 1), b = (−2, 3).
a − b = (5, −2): five right, two down.

(a) On the grid, draw and label the vector **a** – **b**

(1)

p and **q** are vectors such that

$$\mathbf{p} = \begin{pmatrix} 3 \\ c \end{pmatrix} \text{ and } \mathbf{q} = \begin{pmatrix} d \\ -2 \end{pmatrix}$$

Given that $2\mathbf{p} - 3\mathbf{q} = \begin{pmatrix} -9 \\ 14 \end{pmatrix}$

(b) find the value of c and the value of d.

Horizontal: 2(3) – 3d = –9

6 – 3d = –9; –3d = –15; d = 5

Vertical: 2c – 3(–2) = 14

2c + 6 = 14; 2c = 8; c = 4

$c =$ 4

$d =$ 5

(3)

3 marks total
Equate matching components; solve both equations.

(Total for Question 10 is 4 marks)

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